Cancer in Iowa

An evidence review of the burden, contributing risks, prevention and care

Research cutoff: October 22, 2026

What the evidence says

Iowa has a substantial, unusually high cancer burden. The newest live federal tables show an age-adjusted incidence rate about **11.2% above the U.S. rate**. The excess persists after adjusting for age, so an older population alone does not explain it. The death-rate excess is smaller, about **3.5%**, and incidence and mortality rankings should not be confused. [1, 2]

Several contributors are established; a single explanation for the statewide excess is not. Tobacco, alcohol, excess body fat, ultraviolet radiation, certain infections and radon cause cancer. Evidence also warrants attention to particular pesticides, drinking-water contaminants and occupational exposures. Their contributions differ by cancer type, dose, timing and population. [12-15, 19, 22-31, 35-49]

The September 2026 drivers study needs a careful denominator. The Iowa Medical Society's summary says evaluated demographic and risk factors statistically accounted for about 76% of excess cases among the **five modeled cancer types**. That is not 76% of every Iowa cancer diagnosis, and it does not settle the remaining environmental questions. [9, 10]

There are useful actions now. Tobacco cessation, less alcohol, UV protection, indicated vaccination, home radon testing, appropriate well testing, recommended screening and reliable follow-up all belong in Iowa's response. Access to care and pollution prevention matter alongside personal choices. [13-15, 19, 22, 24, 33, 50-60]

The Iowa registry projects **21,700 new diagnoses and 6,400 cancer deaths in 2026**. These are forecasts, not completed 2026 counts. A person can have more than one reportable primary cancer. [3, 4, 8]

How this compilation is organized

The report covers current statistics; cancer types; age, race and geography; the drivers study; established risks; radon, water and agricultural exposures; screening and care; current policy; and research gaps. Numbered citations link to sources. The project also contains all 99 county records, demographic and cancer-type comparisons, original federal downloads, detailed research memos and a source index.

This is a broad evidence review, not an exhaustive systematic review or an assessment of any individual's cancer. Prevention lowers risk; it cannot guarantee that a person will avoid cancer or identify who is responsible for a diagnosis.

1. The current statistical picture

The live National Cancer Institute State Cancer Profiles downloads retrieved October 22 are newer than many search previews and the figures used in the annual Iowa report. [1, 2]

Measure	Iowa	United States	Period
New cancers per 100,000/year	504.1	453.4	2019-2023
Cancer deaths per 100,000/year	148.2	143.2	2020-2024
Recent incidence change/year	+0.7%	-0.6%	Latest modeled 5-year trend
Recent mortality change/year	-0.5%	-1.4%	Latest modeled 5-year trend

All rates are age-adjusted. Iowa incidence has a 95% confidence interval of 500.9-507.3; mortality 146.5-149.8. Iowa's incidence trend is statistically rising. The current combined-sex mortality trend is classified **stable**, because its confidence interval (-1.3% to +0.1%) includes zero; a negative point estimate alone does not establish a decline. Longer historical declines and subgroup trends can differ from this latest combined estimate. [1, 2]

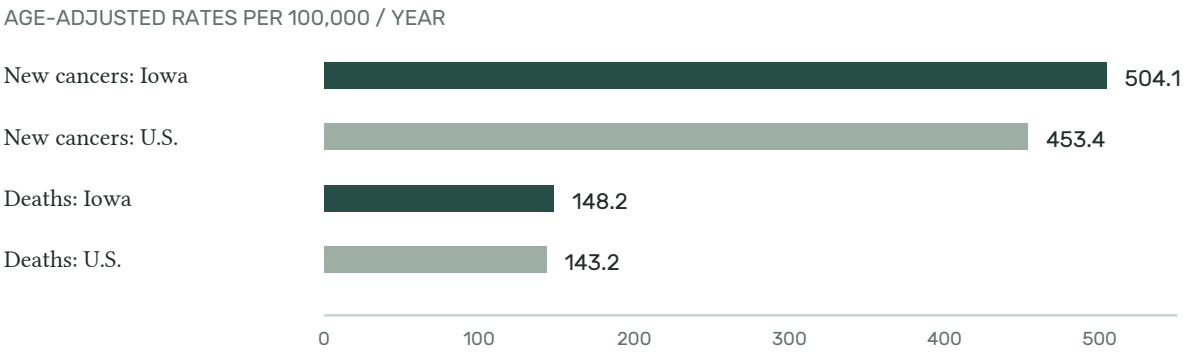
Iowa's rank depends on the table

Older reports describe Iowa as second-highest for incidence. In the current federal download, Iowa's assigned **CI*Rank is 3, with a confidence range of 2-3**; its displayed rate sorts just above West Virginia. The footer explains that displayed SEER rates and the uniform ranking calculation use different numbers of age groups. Both states' rates are close. Iowa's mortality CI*Rank is 22 (confidence range 17-28). [1, 2]

The safe conclusion is that Iowa remains among the highest-incidence states. An unqualified claim that it is the only rising state or the fastest-rising state is not supported by this current table. [1]

Read the measures correctly

Incidence measures new cancers; mortality measures deaths. Age adjustment supports population comparisons but does not remove smoking, screening, racial composition, economic circumstances or environmental differences. Surveillance generally excludes common basal/squamous skin cancers and most in-situ tumors; bladder in-situ is included. A rate of 504.1 per 100,000 is not a personalized annual probability. Dividing the death rate by the incidence rate would not produce a valid survival estimate: the periods and people differ. [1, 2, 8]



2. Which cancers stand out

These are observed age-adjusted incidence rates for 2019-2023, drawn from the saved cancer-specific federal tables. Breast rates use the female population and prostate rates the male population; the others below use both sexes. Each cancer's query URL and confidence interval are retained in the accompanying CSV. [7]

Cancer	Iowa	U.S.	Iowa recent trend
Female breast	139.2	133.0	Rising
Prostate	133.4	119.9	Rising
Lung and bronchus	59.6	51.8	Falling
Colon and rectum	40.3	36.9	Stable
Melanoma	33.7	23.8	Rising
Kidney and renal pelvis	21.6	17.9	Rising
Non-Hodgkin lymphoma	21.4	18.5	Stable
Bladder	21.2	18.7	Falling
Leukemia	17.3	14.5	Rising
Oral cavity and pharynx	14.9	12.3	Rising

Melanoma is roughly 42% higher than the U.S. point rate. Lung incidence is declining in Iowa, but more slowly than nationally. The excess is not confined to the five cancers most often discussed in headlines. Leukemia, lymphoma and kidney cancer also warrant attention. These categories contain biologically different subtypes, so a statewide rate cannot identify one shared exposure. [7]

Diagnoses and deaths tell different stories

The annual registry report projects 2026 diagnoses of prostate 3,100; breast 3,050; lung 2,600; colorectal 1,630; and melanoma 1,500. Together these are about 55% of projected new diagnoses. Lung remains the largest projected source of cancer death: 1,460, approximately 23% of 6,400 total deaths. Colorectal and pancreatic deaths are projected at 550 and 530. [3]

A high diagnosis count does not necessarily mean a high death rate. For example, observed 2020-2024 female breast mortality is lower in Iowa than nationally (17.1 versus 18.9), despite higher incidence; lung mortality remains higher (33.7 versus 30.2). Screening, tumor biology and treatment can affect these measures differently. [7]

3. Age, sex and racial differences

Age and younger-onset cancer

Most Iowa cancers occur in older adults. In 2019-2023 federal data, incidence among people under 50 is 116.7 per 100,000 versus 107.6 nationally; the Iowa trend for this broad group is classified stable. For ages 50+, Iowa's rate is 1,518.5 versus 1,363.3, with a rising trend. In 2020-2024, under-50 mortality is 13.8 versus 14.2 nationally and is falling. These are different populations from the commonly used ages 20-39 or 15-39 definitions of young adulthood. [7]

The 2026 registry report's older, clearly defined 2018-2022 analysis recorded 497 diagnoses at ages 0-14, 245 at ages 15-19 and 4,006 at ages 20-39 across five years. Its childhood/adolescent comparison found incidence similar to the U.S.; the ages 20-39 incidence comparison was elevated. These findings should not be generalized into a claim that every pediatric or young-adult cancer is surging. [3]

Men and women

Observed 2019-2023 incidence is higher in Iowa for both men (549.4 versus U.S. 490.4) and women (473.0 versus 428.9). Sex-specific rates reflect different organ sites, exposures and detection patterns. [7]

Racial and ethnic disparities

Current tables show non-Hispanic Black Iowans with incidence of 553.5 versus 458.1 among non-Hispanic Black people nationally. In 2020-2024, cancer mortality is 190.9 among non-Hispanic Black Iowans versus 149.7 among non-Hispanic White Iowans, about 28% higher by the point estimates. Smaller populations produce wider uncertainty; the detailed CSV retains confidence intervals and the original group definitions. [7]

These disparities cannot be interpreted as evidence of an inherent biological cause. Insurance, discrimination, income, exposure, stage at diagnosis, screening and treatment access may contribute in different ways. An all-population state average can conceal these needs. [61, 69]

Inherited susceptibility

NCI estimates that harmful inherited variants account for about 5-10% of cancers. Multiple relatives with related cancers, unusually early diagnoses, ovarian or male breast cancer, or several primary cancers in one person can justify a genetic-risk discussion. Tumor biomarker testing and testing for inherited susceptibility are different services. A negative inherited-risk test does not eliminate cancer risk. [21]

4. Geography: statewide burden, local differences

The project includes a complete 99-county comparison with rates, confidence intervals, annual counts, trend estimates and uncertainty around ranks. The rates below use different periods for diagnoses and deaths and should be read in separate columns. [5, 6]

County	Incidence 2019-2023	Mortality 2020-2024
Cass	601.4	167.4
Pocahontas	591.0	161.7
Palo Alto	563.1	186.8
Marion	532.3	163.5
Mahaska	506.3	173.3
Jasper	504.3	155.8
Iowa	504.1	148.2

All rates are age-adjusted per 100,000/year. Cass's incidence confidence interval is 550.6-656.1 and its rank confidence interval spans 1-35. Palo Alto's mortality rank spans 1-78. Small-county point rankings are imprecise, even where a high burden is clear. Marion, Mahaska and Jasper provide local context around the Pella region; they are not a Pella-specific exposure assessment. [5, 6]

Rural and urban Iowa

Direct federal rural/urban filters show incidence 507.2 in rural Iowa and 501.8 in urban Iowa; mortality is 152.0 versus 145.5. These aggregate rates are preferable to averaging county rates. The statewide problem affects both settings, while rural care barriers deserve particular attention. [7, 69]

What a map cannot show

Registry geography generally means residence at diagnosis. It does not reconstruct earlier homes, drinking-water sources, workplaces or decades of exposure. The same county can contain multiple public-water systems, private wells and different industries. A nearby river's nitrate measurement is not the dose in every resident's tap water. [8, 32, 33]

CDC defines a cluster using an excess of the same or etiologically related cancers in a defined population, area and period. Many cancer types, many counties and flexible time windows create opportunities for chance findings. A concerning pattern can warrant investigation without proving a common environmental cause. Exposure-specific boundaries, latency and residential histories are essential. [64]

5. What the 2026 drivers study adds

Iowa HHS and the University of Iowa launched the Key Drivers of Cancer in Iowa project with a $1 million appropriation. Initial work addressed demographic and behavioral factors, geographic patterns, screening practices and potential interventions. The original scope included six cancer-site groupings; the widely reported percentage concerns five modeled cancer types. [10, 11, 63]

The 76% finding

The accessible Iowa Medical Society summary of the September 2026 Year 1 findings reports that evaluated factors statistically accounted for approximately **76% of excess burden among the five modeled cancers**. Variables included demographic characteristics, tobacco, obesity, alcohol, insurance and selected radon/UV measures. It reports no remaining modeled excess lung cases among adults 50+, only two excess melanoma cases, and residual excess prostate, postmenopausal breast and colorectal cancers. [9]

This is population modeling. It does not show that a particular person's cancer was caused by drinking, weight or smoking. It does not mean 76% of all Iowa diagnoses were preventable, nor that the remaining 24% were caused by pollution. Demographics are not all modifiable, and a model's reduction in residual excess is not automatically a causal attributable fraction. [9] (statistical interpretation)

Environmental questions remain open

The summary describes further environmental investigation as a next phase. County-level risk measures cannot substitute for individual exposure histories; correlated predictors and imperfect measurement complicate interpretation. A lack of a consistent geographic pattern across modeled cancer types does not prove that no environmental exposures matter. [9, 11]

Access limitation

HHS links the Year 1 PDF, but direct access returned HTTP 403 during this research. The final PDF's full methods were **not independently reviewed** here. The September finding is therefore attributed to the accessible clinical summary, with the original report link preserved for follow-up. The February interim report supplied background, but it is not silently substituted for the final report. No new claim that the modeling has undergone independent peer review is made. [9-11]

The useful implication is that Iowa can act on established preventable risks while continuing environmental and genetic research. The two tasks support each other.

6. Established risks relevant to Iowa

Tobacco and alcohol

Tobacco causes cancers well beyond the lung, including bladder, pancreas, kidney and several digestive and head/neck sites. Current smoking prevalence cannot capture the effects of smoking decades earlier. Quitting helps even after a cancer diagnosis. [13]

Alcohol causally increases risk at at least seven cancer sites, including breast and colorectal cancer. Beer, wine and spirits all contribute alcohol exposure. Some risks increase at low levels of consumption; less alcohol means less alcohol-related risk. A case arising at an alcohol-associated site is not automatically alcohol-caused. [14]

Weight, activity and UV

Excess body fat is associated with at least 13 cancer types, including endometrial, colorectal, kidney and postmenopausal breast cancer. Physical activity and a healthy dietary pattern support prevention; no particular food, supplement or weight-loss drug is established as universal cancer protection. [15]

Ultraviolet radiation from sunlight and indoor tanning is a major preventable cause of melanoma. Outdoor work, sunburns and skin susceptibility are relevant in Iowa. Shade, clothing, sunscreen and avoiding tanning beds reduce UV exposure. A high melanoma rate is not, by itself, evidence of contaminated drinking water. [18]

Iowa measurements

Risk measure	Latest Iowa estimate	Population/data year
Current cigarette smoking	11.7%	Adults, 2025 BRFSS
Past-month binge drinking	17.8%	Adults, 2025 BRFSS
BMI at least 30	36.9%	Adults, 2025 BRFSS
HPV vaccination up to date	60.4%	Ages 13-17, 2025 NIS-Teen

The BRFSS brief was published in October 2026. These are weighted crude self-reports. Do not directly trend them against older age-adjusted comparisons. HPV completion nationally was 63.4%; Iowa's confidence interval of 54.4-66.0 cautions against declaring a statistically significant gap from point estimates alone. [16, 20]

A national model attributed 40% of 2019 cancers among adults 30+ to evaluated modifiable factors. That estimate is not an Iowa-specific explanation of its excess and does not assign responsibility to patients. [12]

7. Radon: strong evidence, direct action

Radon is a naturally occurring radioactive gas that can accumulate indoors. It causes lung cancer; smoking and radon together are particularly hazardous. Iowa's geology creates substantial exposure potential. [22-24]

Iowa HHS cites a survey in which **71.6% of tested homes exceeded 4 pCi/L** and places the whole state in EPA Zone 1. This is a historical survey finding, not a 2026 census of every house. It does not establish the level in any particular home. [22]

The Iowa Radon Lung Cancer Study provided individual-level evidence from 413 female lung-cancer cases and 614 controls with long residence in their homes. Its exposure assessment and smoking adjustment make it more informative than simply comparing county radon maps with county lung-cancer maps. Like any study, its estimates have uncertainty and population limits. [23]

Practical response

- Test the home using a suitable protocol, regardless of a neighbor's result, building age or county.
- EPA recommends mitigation at **4 pCi/L or higher**, and considering mitigation at **2-4 pCi/L**. There is no known risk-free threshold.
- Confirm results as directed, use a credentialed mitigation professional where needed, and retest after the system is installed. [22, 24]

Iowa policy now

HHS states that passive radon control is required for qualifying new single- and two-family homes beginning **July 1, 2026**. A passive system or new construction is not proof that indoor levels are low; testing still matters. Existing real-estate transactions generally require disclosure of known results, rather than universal mandatory testing and mitigation. [22]

Public schools must test attendance centers by **July 1, 2027**, with subsequent testing at least every five years and applicable mitigation requirements. Childcare requirements are addressed separately in state guidance. [25]

Radon prevention is justified by established evidence even though this review cannot calculate its precise share of Iowa's statewide cancer excess. Testing and mitigation address a present exposure; they cannot retrospectively establish why a person developed cancer.

8. Drinking water: nitrate, wells and uncertainty

The public-water nitrate limit is **10 mg/L as nitrogen (NO3-N)**, about 44.3 mg/L expressed as nitrate ion. These units must not be mixed. The limit chiefly protects against infant methemoglobinemia; compliance does not prove zero long-term cancer risk. An association below the limit also does not establish a new legal threshold. [26, 34]

What Iowa-related studies show

Study/population	Finding	Important limit
Iowa Women's Health Study, ovarian, 2015	At least 2.98 vs at most 0.47 mg/L NO3-N: HR 2.03 (1.22-3.38)	Historical exposure estimates; postmenopausal cohort
Same cohort, bladder, 2016	At least 4 years above 5 mg/L: HR 1.62 (1.06-2.47)	Average-concentration quartile result was not significant
AHS women, ovarian, 2025	Per 5 mg/L: HR 1.15 (0.96-1.39)	Interval includes no association
AHS men, prostate, 2026	Above 10 vs at most 1 mg/L: HR 1.16 (1.01-1.35)	Above-limit contrast; quartile results weaker

HR means a relative hazard under the study model, not an absolute lifetime probability. These different studies and exposure contrasts cannot be pooled informally. The detailed memo also covers a kidney signal with no consistent dose-response trend. NCI reports both positive and null findings across cancer types, including null Iowa Women's Health Study water-nitrate findings for colorectal and pancreatic cancer. [27-31, 35]

Public systems and private wells differ

Iowa DNR's 2025 compliance report, released June 2026, lists **36 nitrate violations at 13 public systems**. It reports 97.36% of public systems met health-based standards. The violations are not 36 towns or a statewide cancer count. Source rivers and treated finished water are different measurements. [32]

Private wells are not regulated like public supplies. DNR recommends annual nitrate/bacteria testing, with arsenic and manganese testing under its more detailed program guidance every three years. County environmental health can arrange funded testing. A deep or clear-looking well is not proof of safe water. [33]

Boiling does not remove nitrate and can concentrate it. Use a safe alternate source when the limit is exceeded and seek contaminant-specific treatment advice. Proper reverse osmosis or distillation can reduce nitrate; an ordinary carbon pitcher should not be assumed to do so. Verify treatment with testing. [34]

9. Pesticides and farming

Pesticides are many different chemicals. Their cancer evidence cannot be reduced to one label or a total number of pounds applied. Hazard classification asks whether an agent can cause cancer under some conditions; actual risk also depends on exposure, route, duration and susceptibility. [36-40, 48]

Glyphosate

IARC classifies glyphosate as probably carcinogenic (Group 2A). [73] The large 2018 prospective Agricultural Health Study analysis found no association with overall cancer, solid tumors or non-Hodgkin lymphoma. An acute myeloid leukemia signal was imprecise in the main analysis, with a significant result in one lagged analysis; confirmation is needed. [37]

EPA has retained its prior conclusion that glyphosate is "not likely to be carcinogenic to humans," while acknowledging the court's vacatur of the human-health portion and withdrawal of the 2020 interim decision. Its revised assessment was anticipated for late 2026; this review did not find a completed replacement as of October 22. Neither an old approval nor a lawsuit verdict resolves every scientific question. [38]

Other ingredients

IARC upgraded **atrazine and alachlor to Group 2A in November 2025**. Older summaries listing atrazine as unclassifiable are outdated. An AHS atrazine study found a lung-cancer association and selected prostate/lymphoma subgroup signals, with uncertainty and multiple-comparison concerns. Dicamba research identified a liver/bile-duct association while earlier lung/colon signals did not persist with longer follow-up. [36, 39, 40]

Why farmers' lower total incidence does not settle safety

The 2026 registry analysis found approximately 13% fewer cancers overall than expected among Iowa farmers in the AHS, after age/sex adjustment. Lower smoking and other differences can lower total cancer while particular cancers or exposure groups face higher risk. This is not an experiment showing that pesticides protect against cancer, nor proof that nearby nonfarmers are protected. [4]

Animal feeding operations and county studies

A 2026 ecological study found higher cancer incidence in high-density Iowa AFO counties; an individual-level Iowa study found selected blood/lymph associations without a consistent exposure-response pattern. These findings merit further work but do not establish a fixed personal risk from distance to a hog barn or identify the responsible agent. County pesticide maps have similar limits. [41, 42]

Exposure controls remain sensible: follow label-required protection, reduce drift, prevent take-home contamination, use trained application and maintain safe storage. An exact statewide attributable fraction is not required before reducing avoidable exposure.

10. PFAS, industry and workplace risks

PFAS

IARC classifies **PFOA as carcinogenic (Group 1)** and **PFOS as possibly carcinogenic (Group 2B)**. PFAS are a large chemical family; these classifications cannot be assigned indiscriminately to every member. A hazard classification is not a claim that any detection causes cancer. [43]

Iowa's targeted 2021-2022 public-water study found at least one PFAS in 52 of 126 finished-water samples. Sites were selected for potential susceptibility, so this is not a random statewide estimate or a list of current violations. Historical health advisories and later enforceable limits are different benchmarks. [44]

The 2024 federal rule set PFOA and PFOS limits of **4 parts per trillion each**. EPA's May 2026 compliance-extension proposal and changes proposed for other PFAS provisions should not be described as final merely because they were announced. Check the current rule and the water system's actual results. [45]

For a household concern, distinguish a public-system report from private-well testing. Suitable certified filters can reduce specified PFAS, but products, maintenance and replacement schedules matter. No reviewed study establishes a defensible percentage of Iowa's statewide cancer excess caused by PFAS. [46, 47]

Occupational and industrial exposures

Asbestos, benzene, diesel exhaust, respirable silica and other recognized carcinogens are relevant to particular construction, manufacturing, transportation and maintenance jobs. Inorganic arsenic can occur naturally in groundwater. Occupation and geology must be measured, rather than inferred from a state ranking. [31, 33, 48]

IARC classifies occupational firefighting as carcinogenic, with sufficient human evidence for mesothelioma and bladder cancer. This supports workplace exposure reduction and surveillance; it is not an estimate for Iowa's general population. [49]

An emissions inventory can identify facilities and chemicals for investigation. Reported pounds released are not absorbed dose, and proximity is not a verified cause of cancer. A useful site assessment needs the contaminant, historical pathway, actual exposure and relevant time period. [8, 64]

The IEC/Harkin 2026 environmental report is archived in this project as a useful literature and policy synthesis. It does not itself provide a new statewide exposure-linked causal estimate. Its claims are best evaluated alongside the original studies, including null and conflicting results. [67]

11. Prevention priorities

The evidence supports acting on several fronts. The following priorities combine individual services with community measures; they are not a ranking of blame or a promise to prevent every case.

Priority	Useful action	Iowa connection
Tobacco	Offer cessation treatment and smoke-free environments	Quitline Iowa and clinical care
Alcohol	Reduce consumption; include cancer risk in counseling	Breast, colorectal and other sites
Radon	Test, mitigate elevated levels and retest	High statewide exposure potential
Water	Review finished-water reports; test private wells	Nitrate, arsenic and selected PFAS concerns
UV	Shade, protective clothing, sunscreen; avoid tanning	High melanoma incidence
Vaccination	Complete indicated HPV/HBV prevention	Prevent infection-related cancers
Screening	Use an appropriate test and finish follow-up	Large insurance-related gaps
Workplaces	Control known carcinogen exposures	Jobs and chemicals differ

Sources: [13-15, 18, 19, 22, 24, 26, 33, 48-60, 71]

Vaccination and infection prevention

CDC's current HPV guidance recommends routine vaccination at ages 11-12, with the option to start at 9, catch-up through 26 and shared discussion for some adults 27-45. HPV vaccination prevents new infections; it does not clear an existing HPV infection, treat an established cancer or replace cervical screening. Hepatitis B prevention and hepatitis C detection/treatment also help address liver-cancer risk. [19, 71, 74]

Close the access gap

Iowa's 2024 BRFSS reported recommended colorectal screening among ages 45-75 at **73.1% for insured people versus 33.5% for uninsured people**. The overall estimate was 71.7%. Mammography within two years was 73.7% among women 40+, a denominator that includes people older than the USPSTF 40-74 group. [17]

Making tests available is only part of prevention. A positive stool test needs colonoscopy; an abnormal mammogram needs diagnostic follow-up; a diagnosis needs timely treatment. Transportation, paid leave, language access and navigation affect whether that sequence is completed. [50, 51, 61, 69]

Reducing pollution and exposure at the source protects people who cannot individually choose their air, water or workplace conditions. Household testing is useful but does not replace public monitoring and effective exposure controls.

12. Screening: what is currently recommended

These are **final USPSTF recommendations checked October 22, 2026**, primarily for people without symptoms in the specified risk groups. Family history, prior cancer/polyps, inherited syndromes, immunocompromise and prior radiation can change the plan. Symptoms require diagnostic evaluation regardless of screening age or a recent normal test. [50-54]

Site	Main final USPSTF recommendation
Breast	Mammography every 2 years, ages 40-74. Higher-risk care may differ.
Colorectal	Screen ages 45-75; selectively 76-85. Options include annual FIT or colonoscopy every 10 years, with other accepted strategies. Abnormal stool tests need colonoscopy.
Lung	Annual low-dose CT, ages 50-80, at least 20 pack-years, currently smoking or quit within 15 years; stop after 15 years since quitting or when health/ability/willingness precludes curative surgery.
Cervical	Ages 21-29: Pap every 3 years. Ages 30-65: Pap every 3 years, high-risk HPV every 5 years, or cotesting every 5 years. Stop after 65 only with adequate prior screening and no high-risk history. After total hysterectomy, screening generally is not needed without prior high-grade lesions or cervical cancer.
Prostate	Ages 55-69: individual PSA decision after discussion; against routine PSA screening at 70+.

Sources: [50-54]. One pack-year is an average pack daily for a year. USPSTF's current final cervical and prostate recommendations remain the 2018 versions; updates in progress are not final guidance.

Guidelines and insurance are not identical

The American Cancer Society's lung guideline does not use the USPSTF 15-year-since-quitting cutoff. This can create different eligible populations. A clinical recommendation, a program's enrollment criteria and an insurer's payment rules are separate questions. [55, 72]

More testing is not automatically better

Screening can reduce deaths from selected cancers, but it can also cause false alarms, procedures, complications and detection of tumors that would never become harmful. PSA screening particularly requires a discussion of benefits and harms. [50-54]

Multicancer blood tests remain a separate, evolving area. NCI describes unresolved clinical-benefit and mortality questions. A signal is not a diagnosis and can lead to extensive follow-up. These tests should not replace established screening on the strength of detection-rate claims alone. [68]

13. Getting care and support in Iowa

Free or reduced-cost screening

The detailed HHS eligibility table for **July 2026-June 2027** lists breast screening generally at ages 40-64 and cervical at 21-64 for eligible uninsured/underinsured people at or below 250% of federal poverty guidelines. Exceptions may apply. It lists uninsured colorectal screening at ages 45-75 with no income restriction, and uninsured lung screening at 50-80 with smoking-history and address criteria. [55]

Lung program geography matters: the page excludes Polk, Black Hawk and Scott counties. A test's medical appropriateness does not guarantee program payment. Older HHS summaries have different wording. **Call 866-339-7909 before arranging services to confirm eligibility and coverage.** Screening/diagnostic assistance and treatment funding are separate. [55]

Specialty treatment and trials

University of Iowa's Holden is the state's only NCI-designated Comprehensive Cancer Center. It provides specialized services and clinical trials; NCI designation does not mean every patient must receive all care in Iowa City. Community networks also deliver substantial oncology care. Mission Cancer + Blood became part of UI Health Care in 2025 and has more than 20 community clinics. Services differ by location. [57, 58, 70]

Ask the treatment team about stage, pathology, relevant tumor biomarkers, inherited-risk evaluation, a second opinion and suitable trials. Trial availability depends on cancer type, prior treatment and other eligibility requirements; investigational treatment is not guaranteed to help. [21, 58]

Financial, travel and survivorship support

- **Tobacco support:** Quitline Iowa, 1-800-QUIT-NOW (784-8669). [56]
- **Financial navigation:** UI Health Care cancer financial assistance, 1-866-393-4605. Its 2026 page lists eligibility conditions including income and residency; assistance is individualized. [59]
- **Lodging and support:** American Cancer Society, 1-800-227-2345. Iowa City Hope Lodge offers eligible outpatients/caregivers free lodging, subject to acceptance and availability. [60]
- **Treatment trials:** UI Health Care cancer trial finder; discuss participation with the treating team. [58]

Survivorship includes recurrence surveillance, late treatment effects, fertility and sexual health, mental health, employment, finances and caregiver needs. Palliative care can address symptoms and quality of life during treatment; end-of-life care is another essential part of a complete cancer response. [61]

14. Iowa's response and current policy

State cancer planning

The 2023-2027 Iowa Cancer Plan covers health equity, prevention, screening, diagnosis/treatment, survivorship and end-of-life care. It supplies a framework for action rather than evidence that all its goals have been achieved. The registry's 99 Counties Project provides local education and a route for discussing community concerns. [61, 65]

Research into the excess

The $1 million state-supported drivers project combines registry and risk-factor data, geographic analysis, screening questions and intervention planning. Its Year 1 result is a step in that work, with additional environmental and genetic questions still to address. Reporting should identify which analyses are completed, available in full and independently evaluated. [9-11, 63]

Water investment

IDALS's June 1, 2026 statement describes HF2771 Farm to Faucet as nearly **$320 million over 12 years**, including infrastructure, conservation and monitoring. It includes a $25 million allocation to expand Central Iowa Water Works nitrate-removal infrastructure. This is not a one-year cancer research budget. Investment and installed equipment should ultimately be assessed against measured exposures and health outcomes; announcements are not evidence that cancer rates have fallen. [62]

Radon rules

New qualifying residential construction requirements began July 1, 2026; public-school testing has a July 1, 2027 deadline. These are concrete exposure-prevention measures, with testing and compliance still important. [22, 25]

Screening authorization changes

HF2635 / 2026 Acts chapter 1087 was signed in May 2026. Its cancer-screening prior-authorization exemptions have **January 1, 2027 applicability** for the specified plans/contracts and requests, including relevant Medicaid/Hawki provisions. The act refers to provider-recommended screening under specified guidelines. It does not make every cancer test free, cover every insurer or abolish all authorization requirements immediately. [66]

What would show progress

Track smoking and vaccination, screening completion and diagnostic follow-up, radon testing/mitigation, well and finished-water results, stage at diagnosis, treatment access and cancer-specific mortality. Cancer latency means some benefits take years; process measures should support, rather than substitute for, long-term outcome evaluation. [61, 64]

15. What remains unresolved

Questions worth pursuing

1. How much excess persists for each cancer subtype after stronger measurement of screening, historical exposures, socioeconomic conditions and inherited risk?
2. Can long-term residential, utility, well, occupational and pesticide histories be linked to registry cases with appropriate privacy protection?
3. How do mixtures, exposure timing and susceptibility change risk, especially in groups underrepresented in existing cohorts?
4. Why do particular counties retain excess for particular cancers, and does the pattern persist under prespecified boundaries and independent replication?
5. Which prevention and care investments measurably improve follow-up, stage, quality of life and mortality across Iowa's diverse communities?

These are research priorities synthesized from the evidence gaps, not findings already established by this compilation. [8-11, 27-49, 61, 64, 69]

Common claims the evidence does not justify

- **"It is just Iowa's age."** The main comparisons are age-adjusted.
- **"Iowa's entire cancer excess is caused by farm chemicals."** No reviewed study establishes that statewide causal fraction.
- **"Farmers have lower total cancer, so pesticide exposure is safe."** Overall totals can hide subtype and exposure-specific risk.
- **"Water meeting the nitrate limit has zero cancer risk."** Compliance and long-term epidemiologic uncertainty are different issues.
- **"The 76% finding explains all Iowa cancers."** It concerns excess in five modeled cancer types.
- **"A dark county on a map proves a cluster."** Counts, uncertainty, related cancer types, timing and exposure all matter. [1, 4, 8, 9, 26-42, 64]

Methods and limits of this review

Primary sources were prioritized: registry/federal downloads, original cohort studies, public-health agencies, current final guidelines and enacted legislation. Selected clinical or advocacy summaries are labeled as such. Positive, null and conflicting findings were retained. Observed data, forecasts and model-derived excess are kept separate.

The work is broad rather than a formally registered systematic review. No individual medical records, household measurements or full industrial-site inventory were analyzed. Some university-hosted reports blocked full-text downloads; these access limits are recorded. A questionable pediatric federal record was excluded from interpretation pending clarification. Newer data and policy can supersede this dated compilation.

The source register follows. Detailed memos, query URLs, confidence intervals and all-county data are in the project folder.

Source register

All sources were checked or retrieved on October 22, 2026. Data periods and access limitations are recorded in the report and detailed memos. Citation 7 is a collection of live federal queries; the exact cancer-specific URLs are retained in the cancer-types CSV, with subgroup and rural/urban originals in *research/sources/*.

1. NCI State Cancer Profiles: all-site incidence by state, live 2019-2023 download. Federal surveillance.

2. NCI State Cancer Profiles: all-site mortality by state, live 2020-2024 download. Federal surveillance.

3. Iowa Cancer Registry: Cancer in Iowa 2026, especially pages 3-6. Registry report; indexed content, direct download blocked.

4. University of Iowa: 2026 report and Agricultural Health Study farming findings. University research release.

5. NCI: all-site Iowa county incidence, 2019-2023. Federal surveillance.

6. NCI: all-site Iowa county mortality, 2020-2024. Federal surveillance.

7. NCI query collection: cancer types, demographics and rural/urban filters; exact query URLs in project data. Federal surveillance; archived queries in research/sources and CSVs.

8. Iowa HHS: Cancer Data definitions and limitations. Official surveillance documentation.

9. Iowa Medical Society: Year One Cancer Drivers clinical summary. Clinical summary of research; not original full methods.

10. Iowa HHS: cancer study landing page and Year 1 report link. Official project landing page.

11. University of Iowa: Key Drivers interim report, February 2026. Interim research report; indexed content.

12. Islami et al. 2024: modifiable risk factors and U.S. cancer burden, original model. Original research; U.S. adults 30+, 2019.

13. NCI: tobacco and cancer. Official cancer-risk and cessation guidance.

14. U.S. Surgeon General: Alcohol and Cancer Risk advisory, 2025. Official evidence assessment.

15. NCI: obesity and cancer. Official cancer-risk guidance.

16. Iowa HHS: 2025 BRFSS findings brief, October 2026. Primary survey report; 2025 crude weighted estimates.

17. Iowa HHS: 2024 BRFSS findings brief, October 2025. Primary survey report; 2024 screening measures.

18. Iowa HHS: melanoma data and UV risk. Official cancer-risk guidance.

19. CDC: HPV vaccination considerations. Current official guidance.

20. CDC MMWR: 2025 adolescent vaccination coverage, September 3, 2026, Table 3. Primary survey analysis.

21. NCI: Genetic Testing for Inherited Cancer Risk. Official clinical information.

22. Iowa HHS: Radon Resources, household survey and residential rules. Official state guidance; historical survey.

23. Field et al. 2000: Iowa residential radon and lung cancer study. Original case-control study; doi:10.1093/oxfordjournals.aje.a010153.

24. EPA: radon action level and its meaning. Official action guidance.

25. Iowa HHS: radon in K-12 schools. Official state requirements.

26. Iowa HHS: public-water nitrate results and units. Official water guidance.

27. Inoue-Choi et al. 2015: nitrate and ovarian cancer, Iowa Women's Health Study. Original cohort study; doi:10.1002/ijc.29365.

28. Jones et al. 2016: nitrate and bladder cancer, Iowa Women's Health Study. Original cohort study; doi:10.1289/EHP191.

29. Ammons et al. 2025: water contaminants and ovarian cancer, Agricultural Health Study. Original cohort study; doi:10.1097/EE9.0000000000000382.

30. Spaur et al. 2026: drinking-water nitrate and prostate cancer, Agricultural Health Study. Original cohort study; doi:10.1093/jnci/djaf350.

31. NCI DCEG: drinking-water contaminant research, including positive and null Iowa findings. Official research synthesis.

32. Iowa DNR: 2025 public-water compliance report, June 2026, printed pages 1 and 21. Primary monitoring report.

33. Iowa DNR: private-well testing and grants. Official testing guidance.

34. EPA: nitrate drinking-water FAQs and response to an exceedance. Official health/treatment guidance.

35. Jones et al. 2017: nitrate and kidney cancer, Iowa Women's Health Study. Original cohort study; doi:10.1097/EDE.0000000000000647.

36. IARC: atrazine, alachlor and vinclozolin evaluation, November 21, 2025. Primary hazard evaluation.

37. Andreotti et al. 2018: glyphosate and cancer incidence, Agricultural Health Study. Original cohort study; doi:10.1093/jnci/djx233.

38. EPA: glyphosate assessment and registration-review status. Official regulatory/scientific status.

39. Remigio et al. 2024: atrazine and cancer incidence, Agricultural Health Study. Original cohort study; doi:10.1289/EHP13684.

40. Lerro et al. 2020: dicamba and cancer incidence, Agricultural Health Study. Original cohort study; doi:10.1093/ije/dyaa066.

41. Son et al. 2026: animal feeding operation density and county cancer incidence. Original ecological study; doi:10.1016/j.envres.2026.124298.

42. Jones et al. 2020: proximity to animal feeding operations and cancer in Iowa AHS. Original cohort analysis; exposure proxies and nonmonotonic findings.

43. IARC: PFOA and PFOS carcinogenicity evaluation, 2023. Primary hazard evaluation.

44. Iowa DNR: targeted PFAS public-water sampling, March 2023 report. Primary monitoring report; 2021-2022 sampling.

45. EPA: proposed PFOA/PFOS compliance extension, May 2026. Official proposal; distinguish from final rule.

46. Iowa DNR: PFAS and private wells. Official testing information.

47. EPA: filters certified to reduce PFAS. Official treatment guidance.

48. NCI: cancer-causing substances in the environment. Official hazard information.

49. IARC: occupational firefighting carcinogenicity evaluation, July 2022. Primary hazard evaluation.

50. USPSTF: final breast screening recommendation, April 2024. Final clinical recommendation.

51. USPSTF: final colorectal screening recommendation, May 2021. Final clinical recommendation.

52. USPSTF: final lung screening recommendation, March 2021. Final clinical recommendation.

53. USPSTF: final cervical screening recommendation, August 2018. Final recommendation; update in progress.

54. USPSTF: final prostate screening recommendation, May 2018. Final recommendation; update in progress.

55. Iowa HHS: cancer screening participant eligibility, July 2026-June 2027. Current program eligibility; confirm with coordinator.

56. Quitline Iowa. Direct service resource.

57. NCI: Holden Comprehensive Cancer Center designation. Official cancer-center profile.

58. UI Health Care: cancer clinical trials. Direct treatment/research resource.

59. UI Health Care: cancer financial and billing assistance. Direct financial-assistance resource.

60. American Cancer Society: Hope Lodge. Direct lodging resource; eligibility/availability apply.

61. Iowa Cancer Consortium: 2023-2027 Iowa Cancer Plan. Statewide collaborative plan.

62. IDALS: Farm to Faucet bill signing and funding, June 1, 2026. Official enacted-program announcement.

63. University of Iowa: Cancer Assessment appropriation narrative. Official project scope/budget.

64. CDC: cancer patterns, cluster definition and investigation limits, May 2026. Official investigation guidance.

65. Iowa Cancer Registry: Cancer in Iowa 99 Counties Project. Direct public-education resource.

66. Iowa Acts 2026 chapter 1087, HF2635, sections 7-11, pages 4-7. Enacted law; January 2027 applicability.

67. IEC/Harkin: Environmental Risk Factors and Iowa's Cancer Crisis, March 2026. Secondary literature/policy synthesis; original studies prioritized.

68. NCI: questions and answers about multicancer detection tests. Official screening evidence guidance.

69. Nash et al. 2026: rural Iowa cancer burden and stakeholder care-access study. Original mixed-methods study; doi:10.1007/s10552-026-02166-9.

70. UI Health Care: Mission Cancer + Blood network acquisition, January 2025. Direct health-system announcement.

71. CDC: hepatitis B prevention. Official infection-prevention guidance.

72. American Cancer Society: lung cancer screening guideline. Clinical guideline; differs from USPSTF.

73. IARC: glyphosate classification and questions, 2015. Primary hazard evaluation.

74. CDC: hepatitis C treatment and liver-cancer prevention. Official clinical guidance.